JOSHUA SLAUGHTER

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EDUCATION

University of Edinburgh — Edinburgh, United Kingdom
School of Informatics — MRC Human Genetics Unit
Marshall Scholar
Ph.D. in Informatics — *October 2022 - October 2026*
Advisors: Ava Khamseh and Chris Ponting
Committee: Ajitha Rajan and Caroline Dalton

University of Maryland, Baltimore County (UMBC) — Baltimore, MD, USA
Meyerhoff Scholar and Goldwater Scholar
B.S. Computer Engineering — *August 2018 – May 2022*
Magna Cum Laude — GPA: 3.8/4.0

RESEARCH INTERESTS

Causal Inference, Machine Learning, Semiparametric Statistics, Complex Trait Genomics, Social Genomics, Public Health Policy, Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

HONORS & AWARDS

Marshall Scholarship	2022 – 2025
Wellcome Women's Health: Genes, Data and Advancing Approaches Poster Award	2025
Rhodes Scholarship Finalist	2022
UMBC College of Engineering and Information Technology Student Leadership Award	2022
Goldwater Scholarship	2021 – 2022
MARC U-RISE Scholarship	2020 – 2022
Meyerhoff Scholarship	2018 – 2022
UMBC Undergraduate Research Award	2019

PUBLICATIONS

[1] **Slaughter J**, Roskams-Hieter B, Labayle O, Dibble JJ, Beentjes SV, Khamseh A, Ponting CP. "Seven replicated genomic associations of myalgic encephalomyelitis/chronic fatigue syndrome: a biobank study". *medRxiv preprint*. 2026. doi:10.64898/2026.09.09.26362115

[2] Slaughter C, Slaughter J, **Slaughter J**, Slaughter G. "Smart Materials Enabling Flexible Bioelectronic Sensing for Decentralized Monitoring". *TrAC Trends in Analtyical Chemistry*. 2026. doi.org/10.1016/j.trac.2026.119204

[3] Genetics Delivery Team, ..., **Slaughter J**, ..., Ponting CP. "Initial findings from the DecodeME genome-wide association study of myalgic encephalomyelitis/chronic fatigue syndrome". *medRxiv preprint*. 2025. doi:10.1101/2025.08.06.25333109

[4] Labayle O, **Slaughter J**, Roskams-Hieter B, Tetley-Campbell K, van der Laan M, Ponting CP, Khamseh A, Beentjes SV. "TarGene: A Nextflow pipeline for the estimation of genetic effects on human traits via semi-parametric methods". *Journal of Open Source Software*. doi.org/10.21105/joss.09603

[5] Labayle O, Roskams-Hieter B, **Slaughter J**, Tetley-Campbell K, van der Laan M, Ponting CP, Beentjes SV, Khamseh A. "Semi-parametric efficient estimation of small genetic effects in large-scale population cohorts". *Biostatistics*. doi.org/10.1093/biostatistics/kxaf030.

[6] Fagan ME, Kim D, Settle W, Ferry L, Drew J, Carlson H, **Slaughter J**, Schaferbien J, Tyukavina A, Harris NL, Goldman E, Ordway EM. "The expansion of tree plantations across tropical biomes". *Nature Sustainability*. 2022. doi:10.1038/s41893-022-00904-w

[7] Reid JL, Fagan ME, Lucas J, **Slaughter J**, Zahawi RA. "The ephemerality of secondary forests in southern Costa Rica". *Conservation Letters*. 2019;12(5):e12607. doi:10.1111/conl.12607

RESEARCH EXPERIENCE

University of Edinburgh October 2022 – October 2026
Ph.D. Researcher – Advisors: Ava Khamseh and Chris Ponting

- Developed *TarGene*, a bioinformatics pipeline for semi-parametric estimation of genetic effects on complex traits at genome-wide scale.
- Implemented semi-parametric estimation strategies in the Julia module, *TMLE.jl* for parameter estimation under biased sampling regimes.
- Uncovered and replicated novel genetic loci associated with ME/CFS risk across independent biobank cohorts.
- Provided project design guidance and support for DecodeME, the world's largest ME/CFS study.
- Performed functional analyses to link ME/CFS-associated loci to regulatory functions elucidating disease mechanisms.

Princeton Neuroscience Institute May 2020 – August 2021
Undergraduate Researcher – Advisors: Jonathan Cohen and Coraline Rinn Iordan

- Correlated multi-layer feature activations from pre-trained convolutional neural networks with neuronal activity recorded via fMRI in the human visual stream when performing classification problems.

University of Maryland, Baltimore County February 2019 – May 2022
Undergraduate Researcher – Machine Learning & Signal Processing Laboratory
Advisors: Tülay Adali and Zois Boukouvalas

- Applied independent component analysis to identify fMRI biomarkers for classifying schizophrenia subtypes.

Carnegie Mellon University May 2019 – August 2019
Undergraduate Researcher – REUSE Program, Institute for Software Research
Advisors: Justine Sherry and Srinivasan Seshan

- Differentiated TCP congestion control algorithms using machine learning to classify non-congestive loss patterns.

University of Maryland, Baltimore County June 2017 – May 2019
Undergraduate Researcher – The Earth From Above Laboratory
Advisor: Matthew Fagan

- Analyzed tree plantation expansion and forest fates across tropical biomes using remote sensing data and classification algorithms.

WORK EXPERIENCE

ForKids June 2026 – Present
Health Policy Analyst – Research, Advocacy, and Government Relations
Supervisor: Bjorn Koxvold

- Analyzed the effects of the benefits cliff on recipients of means-standard services.
- Developed policy briefs for Virginia state legislators on implementing strategies for the reduction of uninsurance rates amongst child populations.
- Founded a working group consisting of professors, legislators, Virginia state department leaders, and health policy experts to address child health policy issues that the Virginia is currently facing.

TEACHING EXPERIENCE

University of Edinburgh

Methods for Causal Inference, Teaching Fellow	Spring 2025
Insights Through Data, Teaching Fellow	Fall 2023, Fall 2024
Representing Data, Teaching Fellow	Spring 2024
Ethical Data Futures, Teaching Fellow	Spring 2024
Ethics and Politics of Data, Teaching Fellow	Fall 2023

University of Maryland, Baltimore County

Introduction to Computer Science, Teaching Assistant	Fall 2019 – Fall 2022

SELECTED TALKS

International Association for Chronic Fatigue Syndrome Myalgic Encephalomyelitis (IACFSME), New York City, USA. 2025. *Replicated association of CLYBL with ME/CFS risk*
Adaptive and Neural Computation Workshop, School of Informatics, Edinburgh, UK. 2024. *Uncovering Gene-Deprivation Interaction with Targeted Learning*
Edinburgh Next Generation of Genomics Symposium (ENGoGS), Edinburgh, UK. October 2024. *Targeted Learning-Based GWAS Pipeline Towards Identifying Non-Linear Genetic Effects*

POSTER PRESENTATIONS

[1] **Slaughter J**, Roskams-Hieter B, Labayle O, Samms GL, Beentjes SV, Khamseh A, Ponting CP. "Targeted Learning Reveals Novel Genomic Risk Loci in ME/CFS." Festival of Genomics. Hinxton, UK. January 2025.

[2] **Slaughter J**, Roskams-Hieter B, Labayle O, Samms GL, Beentjes SV, Khamseh A, Ponting CP. "Targeted Learning Reveals Novel Genomic Risk Loci in ME/CFS." Women's Health: Genes, Data and Advancing Approaches. Hinxton, UK. January 2025.

[3] **Slaughter J**, Iordan M, Cohen J, Peterson J. "Toward Using Convolutional Neural Networks to Predict Human Behavior and Neural Representations." Leadership Alliance National Symposium. Hartford, CT. July 2020.

[4] **Slaughter J**, Boukouvalas Z, Adali T. "Towards Detection of Misinformation Through Machine Learning and Latent Variable Methods." Undergraduate Research & Creative Achievement Day. Baltimore, MD. April 2020.

[5] **Slaughter J**, Ware R, Seshan S, Sherry J. "Using Non-Congestive Loss to Differentiate TCP Reno and TCP Westwood." REUSE Poster Symposium. Pittsburgh, PA. August 2019.

[6] **Slaughter J**. "The ephemerality of secondary forests in southern Costa Rica." Undergraduate Research & Creative Achievement Day. Baltimore, MD. April 2019.

[7] **Slaughter J**. "Analysis of Geospatial Factors Relating to Opiate Overdoses for the Community of Carroll County." Capstone Trade-Show & Presentation Night. Westminster, MD. January 2018.

TECHNICAL SKILLS

Programming	Julia, Python, R, Bash, SQL, MATLAB
Bioinformatics	Nextflow, PLINK
Machine Learning	TensorFlow, PyTorch, scikit-learn, MLJ
Software Development	Git, Github Actions, Docker, Singularity, Conda
Writing	LaTeX, Overleaf, R Markdown, Microsoft Office
Other Software	ArcGIS, COMSOL Multiphysics